

Social Engine Phase 2 SQL Analysis Report

1. Objective

The objective of Phase 2 was to load the cleaned Social Engine data into SQL tables and use reproducible queries to analyze trends, engagement, user behaviour, correlations, and data-quality issues.

All results in this report were generated from the SQLite database. The outputs were not manually hardcoded.

2. Database schema

The database contains two tables.

Users table

Column	Description
user_id	Unique user identifier
location	User location
language	User language
account_created	Account creation date
follower_count	Number of followers

Posts table

Column	Description
post_id	Unique post identifier
user_id	User who created the post
platform	Social-media platform
text_content	Post text
timestamp	Standardized post timestamp
likes	Number of likes
shares	Number of shares
comments	Number of comments
timestamp_original	Original timestamp before standardization

The relationship between the tables is:

```
users.user_id = posts.user_id
```

The database contains:

- 1,500 users
- 12,000 cleaned posts

The users and posts tables are kept separate because user information should not be unnecessarily duplicated across every post record.

3. Query 1: Monthly posting trend

SQL file

```
sql/trend_query.sql
```

Question

How did the number of posts change from month to month?

SQL techniques used

- Common Table Expressions using **WITH**
- **GROUP BY**
- **COUNT**
- **LAG**
- Percentage-change calculation
- **NULLIF** to avoid division by zero

Query logic

The first part of the query groups posts by month and counts the number of posts.

The second part uses the **LAG()** window function to retrieve the previous month's post count.

The final part calculates the percentage change:

```
(current month count - previous month count)
/
previous month count
× 100
```

The first month has no previous month, so its percentage change is **NULL**. This is expected.

Main result

Activity increased from:

April 2024: 110 posts
May 2024: 975 posts

This represents a:

786.36% increase

Activity later decreased from:

April 2025: 876 posts
May 2025: 46 posts

This represents a:

94.75% decrease

Interpretation

Monthly activity shows two major structural changes. The first is a sharp increase in May 2024. The second is a sharp decrease in May 2025.

These changes should be treated as activity shifts requiring investigation. They should not automatically be classified as data errors because a large change in volume does not, by itself, prove that the records are invalid.

The 786.36% increase is also influenced by the relatively small April 2024 baseline of 110 posts.

Output screenshot

```
≡ trend_output.txt X
screenshots > ≡ trend_output.txt
1  MONTHLY POSTING TREND
2  -----
3  month | post_count | previous_month_count | percentage_change
4  2024-01 | 110 | None | None
5  2024-02 | 110 | 110 | 0.0
6  2024-03 | 87 | 110 | -20.91
7  2024-04 | 110 | 87 | 26.44
8  2024-05 | 975 | 110 | 786.36
9  2024-06 | 923 | 975 | -5.33
10 2024-07 | 953 | 923 | 3.25
11 2024-08 | 974 | 953 | 2.2
12 2024-09 | 954 | 974 | -2.05
13 2024-10 | 965 | 954 | 1.15
14 2024-11 | 962 | 965 | -0.31
15 2024-12 | 968 | 962 | 0.62
16 2025-01 | 932 | 968 | -3.72
17 2025-02 | 809 | 932 | -13.2
18 2025-03 | 904 | 809 | 11.74
19 2025-04 | 876 | 904 | -3.1
20 2025-05 | 46 | 876 | -94.75
21 2025-06 | 44 | 46 | -4.35
22 2025-07 | 49 | 44 | 11.36
23 2025-08 | 59 | 49 | 20.41
24 2025-09 | 58 | 59 | -1.69
25 2025-10 | 35 | 58 | -39.66
26 2025-11 | 52 | 35 | 48.57
27 2025-12 | 45 | 52 | -13.46
28
```

4. Query 2: Engagement by platform

SQL file

```
sql/platform_engagement.sql
```

Question

Which platforms have the highest average engagement?

SQL techniques used

- GROUP BY
- COUNT
- AVG

- **ROUND**
- **COALESCE**
- Aggregate comparison

Query logic

The query groups posts by platform and calculates:

- Number of posts
- Average likes
- Average shares
- Average comments
- Average total engagement

The expression below calculates total engagement:

```
COALESCE(likes, 0) + shares + comments
```

For this combined engagement calculation only, missing likes are treated as zero so that posts are not removed from the comparison.

The original missing values in the **likes** column remain missing in the database.

Result

Platform	Posts	Average likes	Average shares	Average comments	Average total engagement
Reddit	2,031	2,487.80	1,002.23	511.18	3,550.43
YouTube	2,073	2,528.33	1,011.83	504.38	3,545.70
Facebook	2,074	2,526.55	984.17	506.94	3,543.78
Instagram	1,989	2,494.51	1,040.84	499.80	3,543.53
Missing	1,784	2,480.82	998.61	496.54	3,492.04
Twitter	2,049	2,439.87	1,005.39	506.13	3,454.84

Interpretation

Reddit has the highest average total engagement per post at:

3,550.43

Twitter has the lowest average total engagement per post at:

3,454.84

The difference is:

$$3,550.43 - 3,454.84 = 95.59$$

The difference is relatively small compared with the overall engagement values. Therefore, platform alone does not explain engagement.

A suitable conclusion is:

Reddit has the highest average total engagement in this dataset, but the platform averages are close. Platform should be treated as one contextual factor rather than the only explanation for engagement.

Important assumption

Missing likes were treated as zero only for the combined engagement calculation. This may underestimate the true engagement of posts where likes were unavailable.

The original missing likes were preserved and were not permanently replaced with zero.

Output screenshot

```
platform_output.txt X
screenshots > platform_output.txt
1  ENGAGEMENT BY PLATFORM
2  -----
3  platform | post_count | average_likes | average_shares | average_comments | average_total_engagement
4  Reddit   | 2031       | 2487.8       | 1002.23        | 511.18           | 3550.43
5  YouTube  | 2073       | 2528.33      | 1011.83        | 504.38           | 3545.7
6  Facebook | 2074       | 2526.55      | 984.17         | 506.94           | 3543.78
7  Instagram | 1989      | 2494.51      | 1040.84        | 499.8            | 3543.53
8  Missing  | 1784       | 2480.82      | 998.61         | 496.54           | 3492.04
9  Twitter  | 2049       | 2439.87      | 1005.39        | 506.13           | 3454.84
10
```

5. Query 3: Top posts within each platform

SQL file

sql/top_posts.sql

Question

Which posts have the highest engagement within each platform?

SQL techniques used

- Common Table Expression using **WITH**
- **ROW_NUMBER**
- **PARTITION BY**
- Ranking within groups
- **COALESCE**

Query logic

The query calculates total engagement for each post:

```
likes + shares + comments
```

Missing likes are treated as zero only for this ranking calculation.

The **ROW_NUMBER()** function ranks posts within each platform:

```
ROW_NUMBER() OVER (  
  PARTITION BY platform  
  ORDER BY total_engagement DESC  
)
```

The **PARTITION BY** clause ensures that posts are ranked separately for each platform.

The query returns the five highest-engagement posts within every platform category.

Result summary

The output contains five ranked posts for each platform category:

- Facebook
- Instagram
- Missing
- Reddit
- Twitter
- YouTube

The highest visible result was an Instagram post with total engagement of:

```
7,893
```

Interpretation

Ranking posts separately within each platform provides a fairer comparison than creating one overall ranking.

Without **PARTITION BY**, the results could be dominated by one platform. With **PARTITION BY**, each platform receives its own ranking.

The output includes the **Missing** platform category. This is intentional because posts without platform information should not be silently removed.

The result is generated dynamically from the database. It is not hardcoded.

Limitation

A high-performing individual post does not prove that its platform has the highest average engagement.

The top-post ranking and the platform-average query answer different questions:

- The ranking query finds individual high-performing posts.
- The platform query compares average platform performance.

Output screenshot

top_posts_output.txt

screenshots > top_posts_output.txt

```

1  TOP 5 POSTS WITHIN EACH PLATFORM
2  -----
3  platform | platform_rank | post_id | user_id | timestamp | likes | shares | comments | total_engagement
4  Facebook | 1 | gmoeib832zbs | user_pe5ycky | 2025-01-24 10:39:09 | 4902.0 | 1880 | 982 | 7764.0
5  Facebook | 2 | a1kiwl618kzy | user_aaari8o | 2025-04-09 00:00:00 | 4811.0 | 1952 | 920 | 7683.0
6  Facebook | 3 | k5qkhdj5zpnw | user_v85ub55d | 2025-03-03 00:00:00 | 4554.0 | 1985 | 938 | 7477.0
7  Facebook | 4 | 5dsm9m39dfj3 | user_6ftbnlhg | 2024-10-21 00:00:00 | 4892.0 | 1884 | 679 | 7455.0
8  Facebook | 5 | 9isy2ek2v4p1 | user_ujllj1n7 | 2025-03-25 13:45:57 | 4646.0 | 1872 | 901 | 7419.0
9  Instagram | 1 | ycjj5zzt7mvx | user_d9971ba6 | 2025-02-19 05:57:35 | 4983.0 | 1919 | 991 | 7893.0
10 Instagram | 2 | pvfl3d8hj7jd | user_csluibwk | 2024-05-15 00:00:00 | 4989.0 | 1840 | 909 | 7738.0
11 Instagram | 3 | tne7s3o4l4wd | user_lr3fagd1 | 2024-10-30 11:45:55 | 4931.0 | 1903 | 878 | 7712.0
12 Instagram | 4 | mb370uy6ajb1 | user_u7cey8cm | 2024-06-07 20:38:21 | 4686.0 | 1970 | 812 | 7468.0
13 Instagram | 5 | ukluqjoekps5 | user_jck2yka1 | 2025-04-07 00:00:00 | 4947.0 | 1619 | 866 | 7432.0
14 Missing | 1 | tdgjyylpua20 | user_8nvzxsuj | 2024-06-02 00:00:00 | 4979.0 | 1932 | 812 | 7723.0
15 Missing | 2 | ji1k7vpa7acq | user_r3r59q7h | 2025-04-15 09:33:12 | 4797.0 | 1980 | 759 | 7536.0
16 Missing | 3 | vkn6dfmf1mk | user_68enpikx | 2024-10-21 08:44:53 | 4674.0 | 1943 | 894 | 7511.0
17 Missing | 4 | dos1fe30wf4a | user_drbochr22 | 2025-03-29 13:39:45 | 4980.0 | 1657 | 770 | 7407.0
18 Missing | 5 | b08e0pljqox5 | user_gq0rypar | 2024-10-22 01:48:34 | 4951.0 | 1781 | 644 | 7376.0
19 Reddit | 1 | wo7py9aljg3t | user_o8le7hqf | 2025-04-28 00:00:00 | 4864.0 | 1981 | 948 | 7793.0
20 Reddit | 2 | g7d0tgdwipoy | user_yhe9m0z0 | 2024-05-13 16:11:16 | 4804.0 | 1842 | 937 | 7583.0
21 Reddit | 3 | d7a3bjdzzi5w | user_24wzfb8b | 2024-09-13 08:18:34 | 4880.0 | 1716 | 904 | 7500.0
22 Reddit | 4 | xzt681si3ibm | user_w4p8zi5g | 2024-09-30 00:14:29 | 4672.0 | 1822 | 985 | 7479.0
23 Reddit | 5 | 3ffnsd3uzxp | user_kk1qdsq3 | 2024-01-07 13:15:59 | 4785.0 | 1923 | 727 | 7435.0
24 Twitter | 1 | fp89qlckn9w | user_h4lueh1i | 2025-03-15 18:19:33 | 4740.0 | 1933 | 955 | 7628.0
25 Twitter | 2 | sohxtia2np4 | user_716l1rz5 | 2024-06-07 05:24:47 | 4654.0 | 1841 | 945 | 7440.0
26 Twitter | 3 | rd1nkejvm94b | user_mzxfajfz | 2025-04-21 00:00:00 | 4846.0 | 1770 | 810 | 7426.0
27 Twitter | 4 | 6po1mmy90ysd | user_wr37s6cm | 2025-01-21 20:30:03 | 4762.0 | 1903 | 720 | 7385.0
28 Twitter | 5 | 076lgd30r7iu | user_ral75hq0 | 2025-04-29 03:50:57 | 4973.0 | 1946 | 465 | 7384.0
29 YouTube | 1 | 5kvuyvf38nqx | user_z0feut2e | 2024-06-09 00:00:00 | 4923.0 | 1971 | 861 | 7755.0
30 YouTube | 2 | 5n161ir5hhhr | user_u98jwp3f | 2025-01-26 00:44:23 | 4751.0 | 1981 | 878 | 7610.0
31 YouTube | 3 | rh7jt9kj7tw8 | user_01mu2x1s | 2025-01-08 01:23:46 | 4881.0 | 1787 | 913 | 7581.0
32 YouTube | 4 | za6nw486cyn2 | user_rj984hbd | 2025-03-04 11:24:45 | 4915.0 | 1748 | 877 | 7540.0
33 YouTube | 5 | 2ubp2yyrtncq | user_q2y6x6ct | 2025-01-08 02:45:12 | 4900.0 | 1602 | 970 | 7472.0
34

```

6. Query 4: User behaviour groups

SQL files

Detailed user-group query:

```
sql/user_behaviour.sql
```

Summary query:

```
sql/user_behaviour_summary.sql
```

Question

Can users be grouped based on posting frequency and average engagement?

SQL techniques used

- `INNER JOIN`
- `COUNT`
- `AVG`
- Common Table Expressions
- `CASE`
- Conditional classification
- `GROUP BY`

Behaviour groups

Group	Meaning
<code>high-frequency_high-impact</code>	Many posts and high average engagement
<code>high-frequency</code>	Many posts but lower average engagement
<code>high-impact</code>	Fewer posts but high average engagement
<code>occasional_or_low-impact</code>	Lower posting frequency and lower average engagement

The thresholds used are:

```
High-frequency: at least 15 posts  
High-impact: average engagement of at least 5,000
```

Result

Behaviour group	Users	Percentage
<code>occasional_or_low-impact</code>	1,459	97.27%
<code>high-frequency</code>	25	1.67%
<code>high-impact</code>	16	1.07%

Behaviour group	Users	Percentage
high-frequency_high-impact	0	0.00%

The percentages total approximately 100%. Minor differences may occur because the values are rounded to two decimal places.

Interpretation

Most users were classified as occasional or low-impact:

1,459 users, or 97.27% of the sample

Only 41 users belonged to either the high-frequency or high-impact categories.

No user met both thresholds simultaneously.

This suggests that high activity and high average engagement are concentrated among a small number of users.

Limitation

The thresholds of 15 posts and 5,000 average engagement are operational definitions created for this analysis.

They are not universal industry standards. Changing the thresholds would change the group sizes.

Output screenshot

```
≡ behaviour_output.txt X
screenshots > ≡ behaviour_output.txt
1  USER BEHAVIOUR SUMMARY
2  -----
3  behaviour_group | user_count | percentage_of_users
4  occasional_or_low-impact | 1459 | 97.27
5  high-frequency | 25 | 1.67
6  high-impact | 16 | 1.07
```

7. Query 5: Follower count and engagement correlation

SQL file

sql/follower_correlation.sql

Question

Is follower count related to average engagement?

SQL techniques used

- Common Table Expressions
- `INNER JOIN`
- `AVG`
- Centered values
- Pearson correlation calculation
- `CROSS JOIN`
- `ROUND`
- `NULLIF`

Result

```
Users in sample: 1,500
Pearson correlation: -0.0363
```

Interpretation

The correlation is very close to zero:

```
-0.0363 ≈ 0
```

This indicates almost no linear relationship between follower count and average engagement in this sample.

A suitable conclusion is:

The Pearson correlation between follower count and average user engagement was -0.0363 across 1,500 users. This indicates almost no linear association in the sample. Therefore, follower count alone does not appear to be a reliable predictor of average engagement.

Limitations

This result measures linear association only.

It does not prove that follower count can never influence engagement.

Other factors may still matter, including:

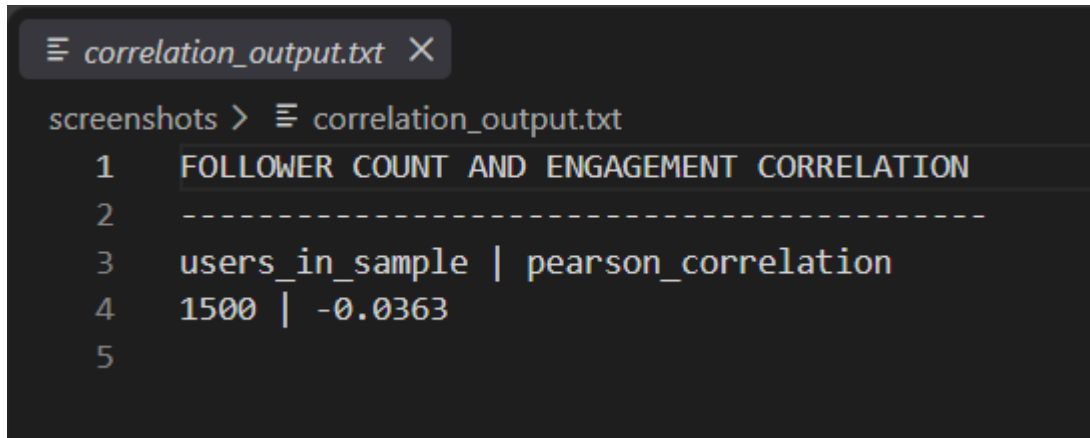
- Platform
- Post topic
- Text content
- Posting time
- Nonlinear relationships

- Missing likes

The engagement calculation also treats missing likes as zero for the combined engagement measure. This may underestimate engagement for posts where likes were unavailable.

Correlation does not prove causation.

Output screenshot



```
correlation_output.txt X
screenshots > correlation_output.txt
1 FOLLOWER COUNT AND ENGAGEMENT CORRELATION
2 -----
3 users_in_sample | pearson_correlation
4 1500 | -0.0363
5
```

8. Query 6: Missingness by platform

SQL file

```
sql/missingness_by_platform.sql
```

Question

Does data quality vary by platform?

SQL techniques used

- GROUP BY
- COUNT
- SUM
- CASE
- COALESCE
- Percentage calculation

Query logic

The query calculates, for each platform:

- Total post count
- Number of missing text values
- Number of missing likes values
- Percentage of missing text

- Percentage of missing likes

The **CASE** expression checks whether a field is missing.

The **SUM** function counts the rows where the condition is true.

Result

Platform	Posts	Missing text	Missing text %	Missing likes	Missing likes %
Twitter	2,049	297	14.49%	417	20.35%
YouTube	2,073	290	13.99%	409	19.73%
Instagram	1,989	277	13.93%	392	19.71%
Missing platform	1,784	230	12.89%	348	19.51%
Facebook	2,074	289	13.93%	389	18.76%
Reddit	2,031	305	15.02%	368	18.12%

Interpretation

Twitter has the highest missing-like rate:

20.35%

Reddit has the lowest missing-like rate:

18.12%

However, missing likes occur across all platform categories. This suggests that missing likes are a broad intake-quality issue rather than a problem isolated to one platform.

Reddit has the highest missing-text rate:

15.02%

There are also 1,784 posts without a platform value. These posts were retained under the separate **Missing** category.

No platform was inferred without evidence.

Output screenshot

```
missingness_output.txt X
screenshots > missingness_output.txt
1 MISSINGNESS BY PLATFORM
2 -----
3 platform | post_count | missing_text | missing_likes | missing_text_percentage | missing_likes_percentage
4 Twitter | 2049 | 297 | 417 | 14.49 | 20.35
5 YouTube | 2073 | 290 | 409 | 13.99 | 19.73
6 Instagram | 1989 | 277 | 392 | 13.93 | 19.71
7 Missing | 1784 | 230 | 348 | 12.89 | 19.51
8 Facebook | 2074 | 289 | 389 | 13.93 | 18.76
9 Reddit | 2031 | 305 | 368 | 15.02 | 18.12
10
```

9. Limitations and assumptions

Missing likes

Missing likes were treated as zero only when calculating combined engagement.

This may underestimate engagement for posts where likes were unavailable.

The original missing values were preserved in the cleaned dataset.

Behaviour thresholds

The thresholds used for behavioural grouping were:

At least 15 posts
At least 5,000 average engagement

These thresholds were created for this analysis and should not be treated as universal industry standards.

Correlation

The Pearson correlation measures linear association only.

It does not prove causation and does not detect every possible type of relationship.

Monthly trends

The monthly trend query identifies changes in activity but does not explain why those changes occurred.

The large percentage increases and decreases are sensitive to the size of the comparison month.

Missing platform values

Posts with missing platforms were retained rather than deleted or assigned to a platform without evidence.

10. Conclusion

The cleaned Social Engine data was successfully loaded into a two-table SQLite database containing 1,500 users and 12,000 posts.

Reproducible SQL queries were used to:

1. Detect monthly posting trends.
2. Compare engagement by platform.
3. Rank top posts within each platform.
4. Group users by activity and impact.
5. Measure the relationship between follower count and engagement.
6. Monitor missing values by platform.

The main findings are:

1. Posting activity changed sharply during the observation period.
2. Reddit had the highest average total engagement, but platform differences were small.
3. High activity and high engagement were concentrated among a small number of users.
4. Follower count had almost no linear relationship with average engagement.
5. Missing likes were common across every platform category.
6. Missing platform values were retained as an explicit data-quality category.

All query results were generated from the SQL database rather than hardcoded.

This report, the SQL files, the database setup script, and the cleaned datasets together provide a reproducible Phase 2 workflow.